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① Water Jug Problem (4-Gallon & 3- Gallon Jugs) :-

· Soln :-

· Step	Action	4-Gallon Jug	3 - Gallon Jug
· Step-1	Fill the 3-Gallon Jug	0	3
· Step - 2	Pour in 4 - Gallon Jug	3	0
· Step - 3	Fill the 3-Gallon Jug	3	3
· Step - 4	Pour in 4-Gallon Jug until its full	4	2
· Step - 5	Empty the 4-Gallon Jug	0	2
. Step - 6	Pour the remaining water into 4-gallon Jug	2	0

· Result :-

→ Exactly 2 gallons of water are obtained in the 4 gallon Jug.

· Interpretation :-

→ It demonstrates state-space search, where each state represents the quantity of water in both Jugs. By applying valid actions systematically, the goal state (2,0) is reached without any measuring device.

(2) MARS ROVER AS AN INTELLIGENT AGENT :-

→ A Mars Rover is an intelligent agent that explores the Martian surface, collects scientific data.

(i) PERCEPTS OF THIS AGENT :-

→ Images of the Martian surface from cameras.

→ Distance & obstacle information from navigation censors.

→ Soil & rock composition from scientific instruments.

(ii) characterize the operating environment :-

→ It operates in a partially observable, stochastic, sequential, dynamic, continuous & single-agent environment. It is partially observable because the rover can't directly know everything about its surrounding.

(iii) What Actions can the agent can take?

→ The Rover can move across the surface, avoid obstacles, capture images, collect soil & rock samples analyze using scientific instruments & transmit collected data to Earth.

(iv) How can its performance be evaluated?

→ Performance can be measured by the number & quality of samples analyzed, scientific discoveries made, distance explored, navigation accuracy, energy efficiency & succesful transmission of data.

(v) Which agent is more suitable? Why?

→ Model-Based, goal-based agent with utility based decision making is suitable. The model based component maintains infos about the environment, while the goal-based component plans routes & tasks.

. Intrepretation :-

→ combining these approaches allows the rover to make informed decisions, navigate safely & maximize scientific discoveries.

(3) EIGHT-QUEENS PROBLEMS USING SEARCH STRATERGIES :-

·Soln :-

·PROBLEM FORMULATION :-

(1) Initial State :- An empty 8 x 8 chessboard.

(2) State space :- All possible arrangements of queens on the chessboard.

(3) Actions :- Place a queen in a valid square, preferbly one queen per column.

(4) Transition model :- Adding a queen to an unoccupied square creates a new board config.

(5) Goal Test :- All 8 queens are placed with no two sharing the same row, column (or) diagonal.

(6) Path Cost :- Each queen placement can be assigned a cost of 1. A complete soln : requires 8 placements.

·Search Strategy :-

→ Start with first column & place a queen in a row

→ Move to the next column & place another queen

→ If no safe position exist, return to previous column & change the queen's position.

→ continue until all 8 placed succesfully.

·One Valid Solution :-

→ The Rover can move across the surface, avoid obstacles, capture images, collect soil & rock samples analyze using scientific instruments & transmit collected data to Earth.

(iv) How can its performance be evaluated?

→ Performance can be measured by the number & quality of samples analyzed, scientific discoveries made, distance explored, navigation accuracy, energy efficiency & succesful transmission of data.

(v) Which agent is more suitable? Why?

→ Model-Based, goal-based agent with utility based decision making is suitable. The model based component maintains infos about the environment, while the goal-based component plans routes & tasks.

· Intrepretation :-

→ Combining these approaches allows the rover to make informed decisions, navigate safely & maximize scientific discoveries.

→ The positions corresponds to column 1 to 8 :-

[1, 5, 8, 6, 6, 3, 7, 2, 4]

→ Means, the queens in each column is placed in the specified row. No two queens can share a row, column (or) diagonal.

• Intrepetation :-

→ Backtracking reduces unnecessary exploration by rejecting invalid configurations early. The search continues until a valid arrangement.

④ OLA Cab Booking Using a Problem-solving Agent :-

• Agent Type :-

→ A goal based problem-solving agent is suitable because the customer wants to reach a specified destination while selecting a cab type.

• Pseudocode :-

```
BEGIN
INPUT pickup_location
INPUT destination
INPUT preferred_cab_type
INPUT budget
INPUT priority // comfort, cost, or time
cab_types = [Mini, Micro, Sedan, Shared, Prime]
```

```
available_cabs = FIND-CABS (
    pickup_location,
    destination,
    cab_types
)
IF available_cabs is empty THEN
    DISPLAY "No cabs Available"
ELSE
    FOR each cab = SELECT_BEST_CAB (
        available_cabs,
        preferred_cab_type,
        budget,
        priority
    )
    DISPLAY suitable_cab
    ASK customer for confirmation
    END IF
    END
```

• Interpretation :-

→ The goal-based agents helps the customer reac the destination while making a suitable choice based on cost, comfort & travel time.